

AI in Genomics 2026

Tunisia · 14–18 Sept 2026

PRACTICAL 01 · DAY 1 · 11:20–12:30

Predict

Submit your sequence data to predict models, interpret and compare scores across complexes.

Presenter: Amal Boukteb

PROGRAMME



01 **Predict**



02 Interpret



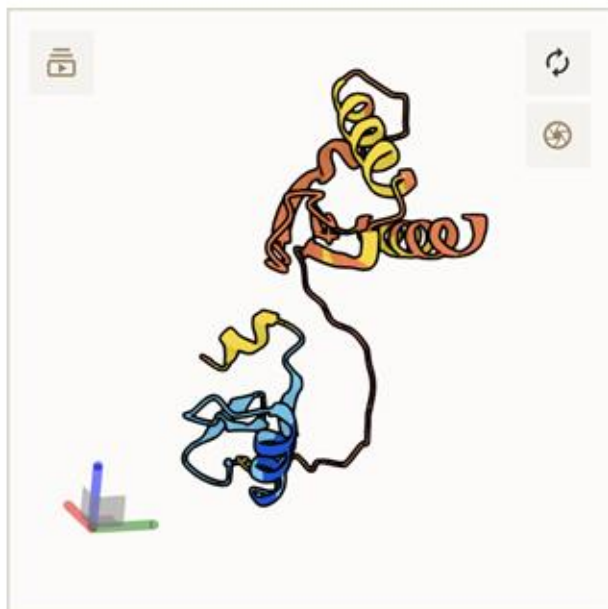
03 Compare



04 Design

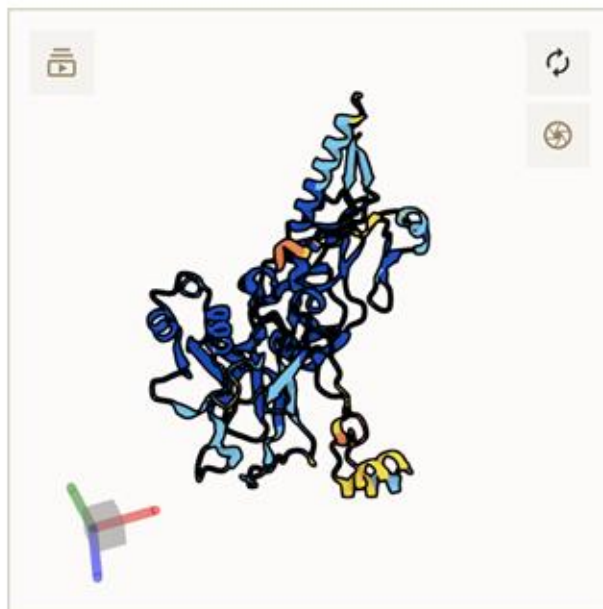
Interpreting AlphaFold
confidence metrics
(recap)

Not all predictions have the same degree of confidence
and it's important to be able to tell them apart



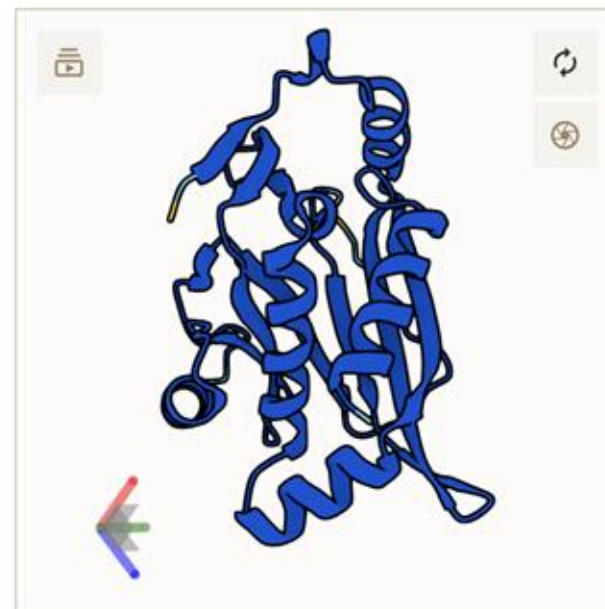
C-type lectin-like domain family
(AF-Q8IZS7-F1)

May function in mediating immune
cell-cell interactions



Apical membrane antigen 1 (AF-
Q3S2X4-F1)

Involved in parasite invasion of
erythrocytes



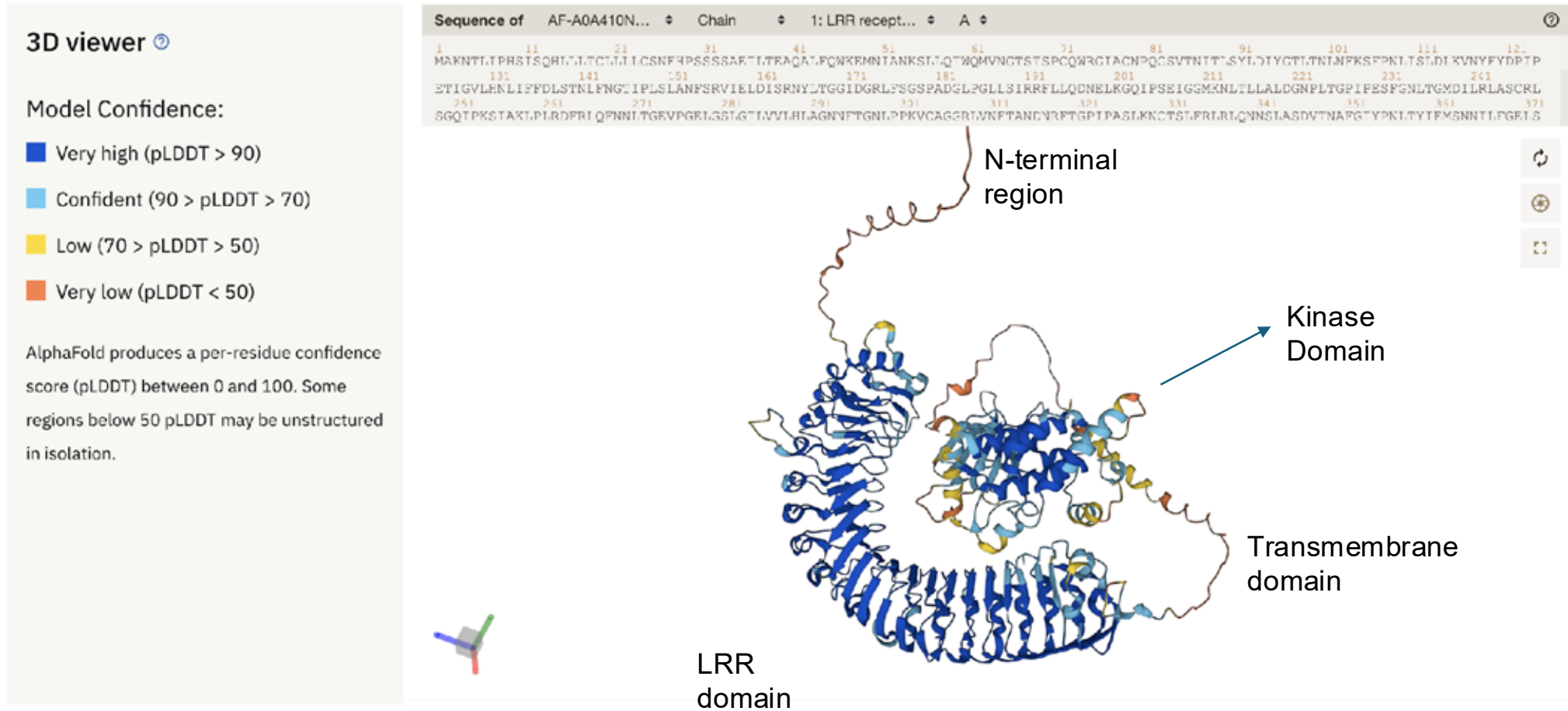
Chloramphenicol
acetyltransferase (AF-P36883-F1)

Responsible for resistance to the
antibiotic chloramphenicol

■ Very high (pLDDT > 90) ■ High (90 > pLDDT > 70) ■ Low (70 > pLDDT > 50) ■ Very low (pLDDT < 50)

Per-residue confidence

Predicted Local Distance Difference Test (pLDDT)



<https://alphafold.ebi.ac.uk/entry/A0A410N673>

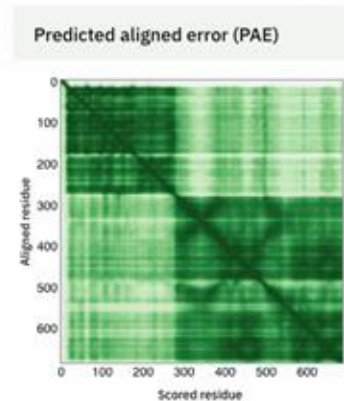
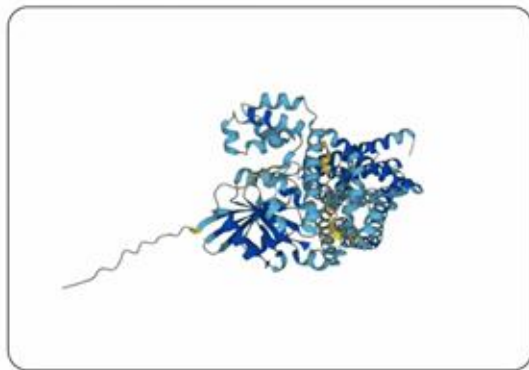
Example: an LRR-RK receptor

AF2 confidence metrics, **PAE**

Predicted alignment error (**PAE**), measures global confidence.

- Measure of the **relative** confidence of the position of **two residues** within the predicted structure.
- PAE is defined as the expected positional error at residue X, in Ångströms (Å), if the predicted and actual structures were aligned on residue Y.

Ferrous iron transport protein B



Model Confidence

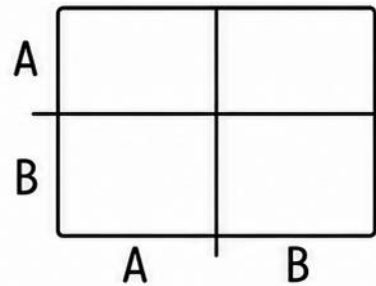


“PAE is effectively a measure of how confident AlphaFold2 is that **domains are well packed**, and that the **relative placement of the domains in the predicted structure is correct**.”

Interpreting PAE plots

1

LOW CONFIDENCE OVERALL

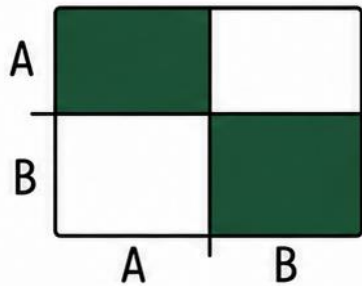


pTM

low

2

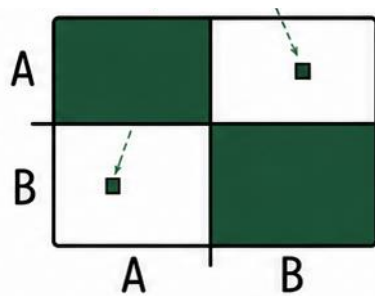
CONFIDENT IN A & B INDIVIDUALLY



high

3

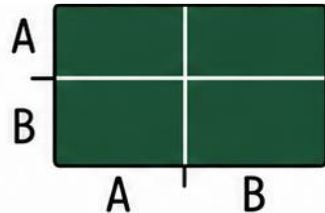
CONFIDENT ONLY AT THE INTERFACE



high

4

FULL RELATIVE CONFIDENCE

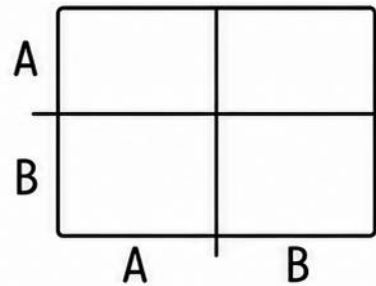


high

Interpreting PAE plots

1

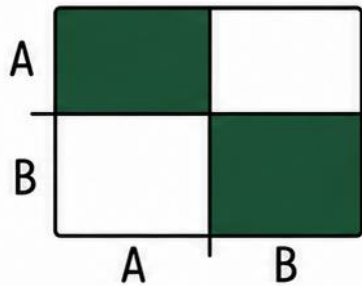
LOW CONFIDENCE OVERALL



pTM low
ipTM low

2

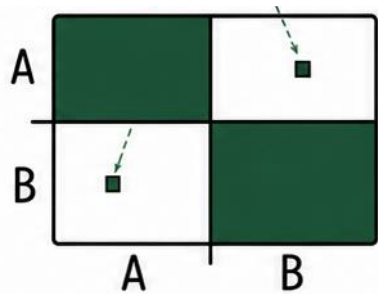
CONFIDENT IN A & B INDIVIDUALLY



high
low

3

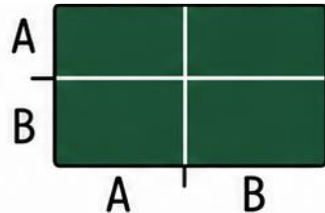
CONFIDENT ONLY AT THE INTERFACE



high
low

4

FULL RELATIVE CONFIDENCE

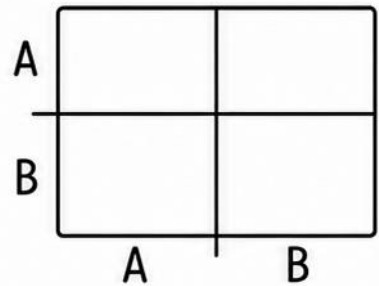


high
high

Interpreting PAE plots

1

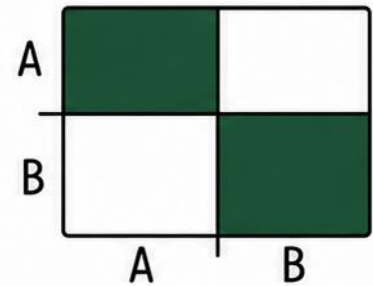
LOW CONFIDENCE OVERALL



pTM low
ipTM low
iPAE low

2

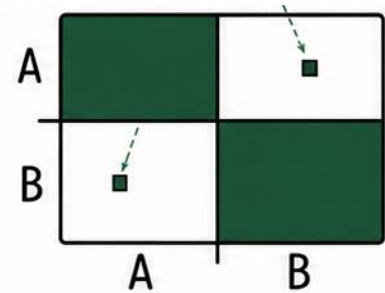
CONFIDENT IN A & B INDIVIDUALLY



high
low
low

3

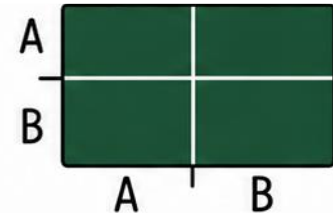
CONFIDENT ONLY AT THE INTERFACE



high
low
high

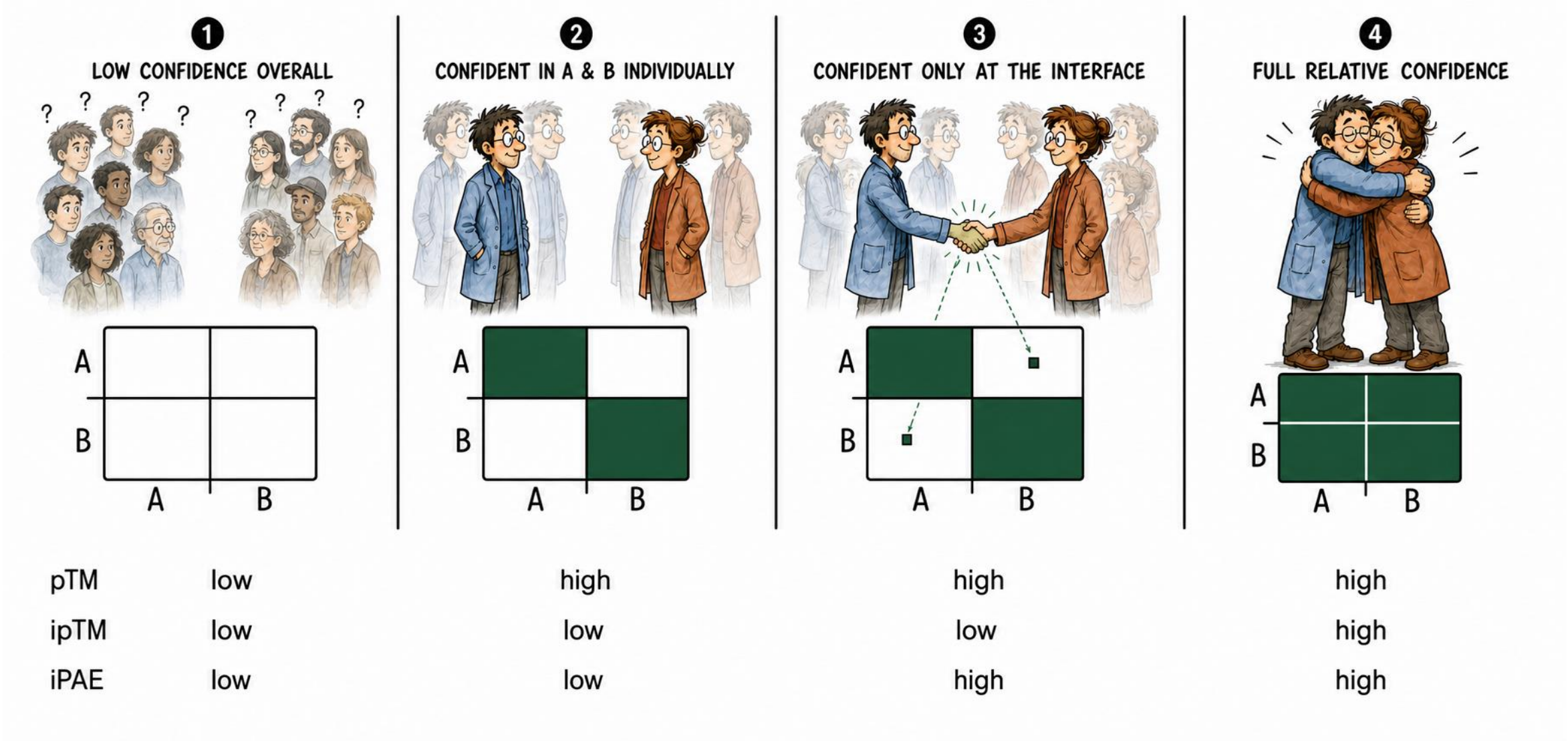
4

FULL RELATIVE CONFIDENCE



high
high
high

Interpreting PAE plots



SESSION 1

Running jobs on the AlphaFold Web Server and interpreting confidence metrics

Rice immune receptor Pikp-1 causes resistance to *Pyricularia oryzae*

Rice blast disease caused by *Pyricularia oryzae*



Pyricularia oryzae
spores



- *Pikp-1*



+ *Pikp-1*

Rice cultivars



Protein sequence of Pikip-1

MEAAAMAVTAATGALAPVLVKLAALLDDGECNLLEGSRSDAEFIRSELEAVHSLLTPNILGRMGD
DDAACKDGLIAEVRELSYDLDDAVDDFLELNFEQRRSASPFGEKARVEEHVSNRFSWKLPAAS
LPPSSVHRRAGLPPPDALVGMDKRMEELTKLLEQGSNDASRWRKRKPHFPLRKTGLKQKIVIKV
AMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQA
NKDVKETTPMLAPVKSICEFHVKVTVCILGLPGGGKTTVARELYDALGTHFPCRVFVSVSPSSSP
SPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEE
WEVIRKSIPKNDLGGRIIMTTRLNSIAEKCHTDDNDVFVYEVGDLNNDALSLSWGIATKSGAGN
RIGTGEDNPCYDIVNMCYGMPLALIWLSSALVGEIEELGGAEVKKCRDLRHIEDGILDIPSLQPL
AESLCLGYNHLPLYLRLLLYCSAYHWSNRIERGRLVRRWIAEGFVSEEKEAEGYFGELINRGWI
TQHGDNNSYNYYEIHVPMLAFLRCKSKEYNFLTCLGLGSDTSTSASSPRLIRRLSLQGGYPVDCL
SSMSMDVSHTCSLVVLGDVARPKGIPFYMFKRLRVLDLEDNKDIQDSHLQGICEQLSLRVRYLGL
KGTRIRKLPQEMRKLKHLEILYVGSTRISELPQEIGELKHLRILDVRNTDITELPLQIRELQHLH
TLDVRNTPISELPPQVGKLQNLKIMCVRSTGVRELPKEIGELNHLQTL DVRNTRVRELPWQAGQI
SQSLRVLAGDSGDGVRLPEGVCEALINGIPGATRAKCREVLSIAIIDRFGPPLVGIFKVPGSHMR
IPKMIKDHFRVLSCLDIRLCHKLEDDDDQKFLAEMP NLQTLVLRFEALPRQPITINGTGFQMLESF
RVDSRVPRIAFHEDAMPNLKLLFEKFYAGPASNDAIGITNLKSLQKVVFRCSPWYKSDAPGISAT
IDVVKKEAEEHPNRPITLLINAGYKEISTESHGSSSENIAGSSGIDTEPAQAQHDNLPAVRDDYKG
KGILLDGRCP TCGRATKIEEETQDRVADIEIQTTTS

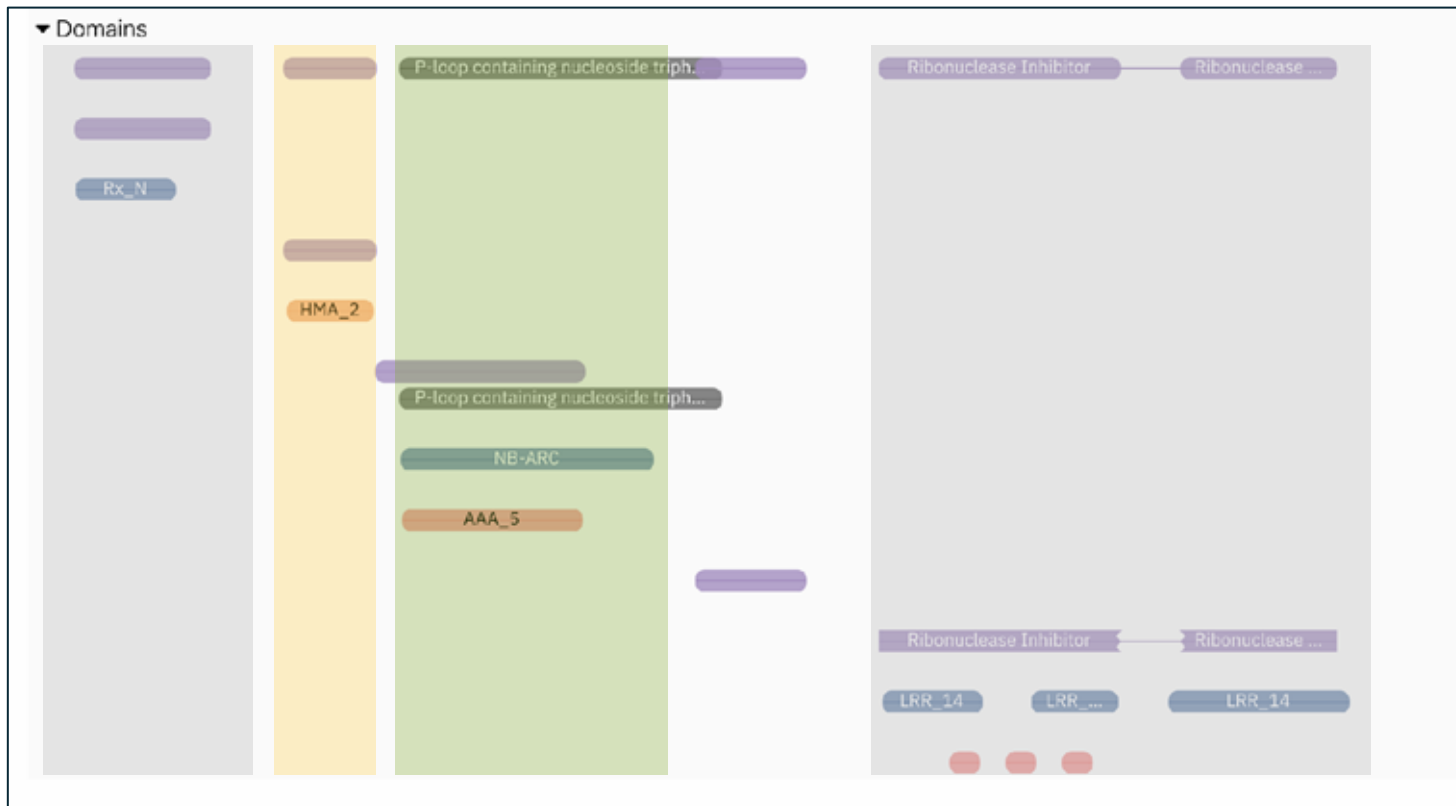
Protein sequence of Pikip-1

MEAAAMAVTAATGALAPVLVKLAALLDDGECNLLEGSRSDAEFIRSELEAVHSLTPNILGRMGD
DDAACKDGLIAEVRELSYDLDDAVDDFLELNFEQRRSASPFGELEKARVEEHVSNRFSDWKLPAAS
LPPSSVHRRAGLPPPDAELVGMDKRMEELTKLLEQGSNDASRWRKRKPHFPLRKTGLKQKIVIKV
AMEGNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQA
NKDVKETTPMLAPVKSICEFHKVKTVCILGLPGGGKTTVARELYDALGTHFPCRVFVSVSPSSSP
SPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEE
WEVIRKSIIPKNDLGGRIIMTTRLNSIAEKCHTDDNDVFVYEVGDLNDNDALSLSWGIATKSGAGN
RIGTGEDNPCYDIVNMCYGMPLALIWLSSALVGEIEELGGAEVKKCRDLRHIEDGILDIPSLQPL
AESLCLGYNHLPLYLRTLILLYCSAYHWSNRIERGRLVRRWIAEGFVSEEKEAEGYFGELINRGWI
TQHGDNNSYNYEIHVPMLAFLRCKSKEYNFLTCLGLGSDTSTSASSPRLIRRLSLQGGYPVDCL
SSMSMDVSHTCSLVVLGDVARPKGIPFYMFKRLRVLDLEDNKDIQDSHLQLOGICEQLSLRVRYLGL
KGTRIRKLPQEMRKLKHLEILYVGSTRISELPQEIGELKHLRILDVRNTDITELPLQIRELQHLH
TLDVRNTPISELPPQVGKLQNLKIMCVRSTGVRELPKEIGELNHLQTL DVRNTRVRELPWQAGQI
SQSLRVLAGDSGDGVRLPEGVCEALINGIPGATRAKCREVLSIAIIDRFGPPLVGIFKVP GSHMR
IPKMIKDHFRVLSCLDIRLCHKLEDDDDQKFLAEMP NLQTLVLRFEALPRQPITINGTGFQMLESF
RVDSRVPRIAFHEDAMPNLKLEFKFYAGPASNDAIGITNLKSLQKVVFRCSPWYKSDAPGISAT
IDVVKKEAEEHPNRPITLLINAGYKEISTESHGSSSENIAGSSGIDTEPAQAQHDNLPAVRDDYKG
KGILLDGRCP TCGRATKIEEETQDRVADIEIQTETTS

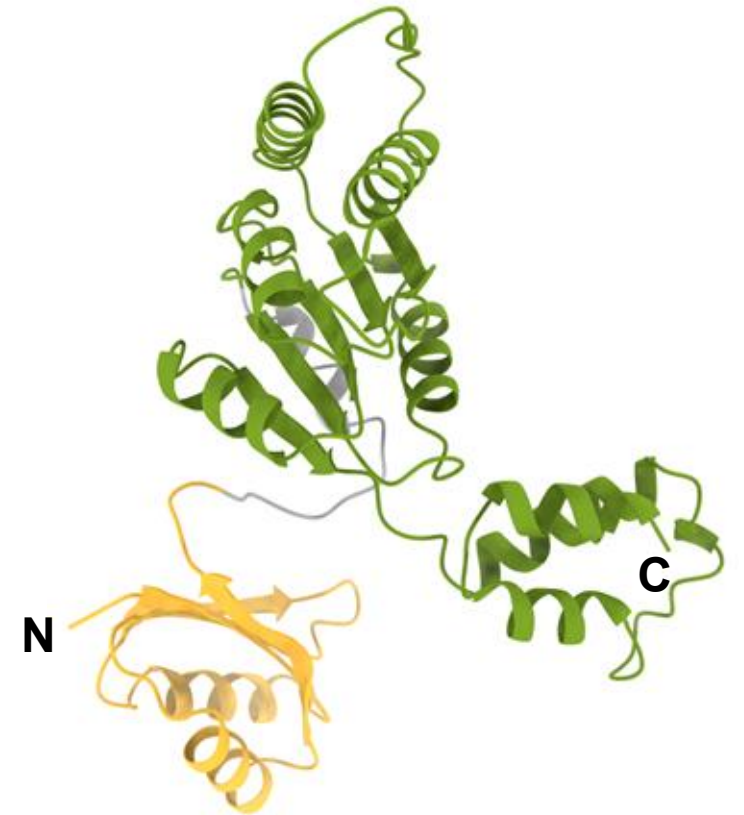
Residues 186–486

Domain architecture of Pikp-1

Residues 186–486



Domain annotation from InterProScan



EXERCISE 1

Modelling individual structures of proteins

In this exercise, you will model the structure of the partial sequence of the rice immune receptor protein, **Pikp-1**.

To begin, navigate to <https://alphafoldserver.com/>, create an account, and submit your first job:

>Pikp-1_186-486

```
GLKQKIVIKVAMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQANKDVKETTPMLAPVKSICEFHKVKTVCILGLPGGGKTTVARELYDAL  
GTHFPCR VFVSVPSSSPSPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEWEVIRKSI PKNDLGGRI IMTTRLNSIAEKCHTDDNDVFV  
YEVGDLNNDALSLSWGIATKSGAGNRIGTGEDNPCYDIVNMCYGMPLALIWLSSAL
```

Modelling individual structures of proteins

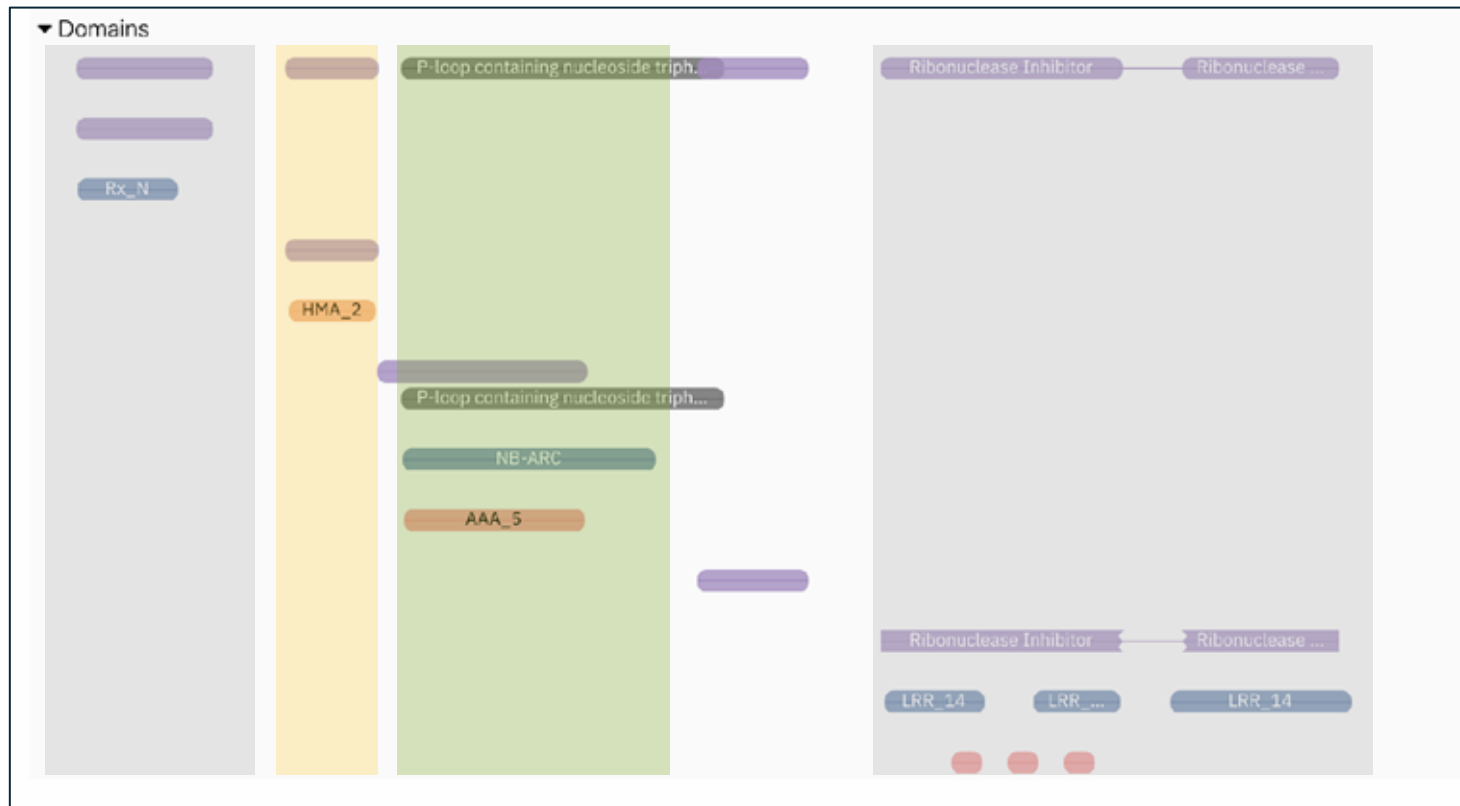


DISCUSSION

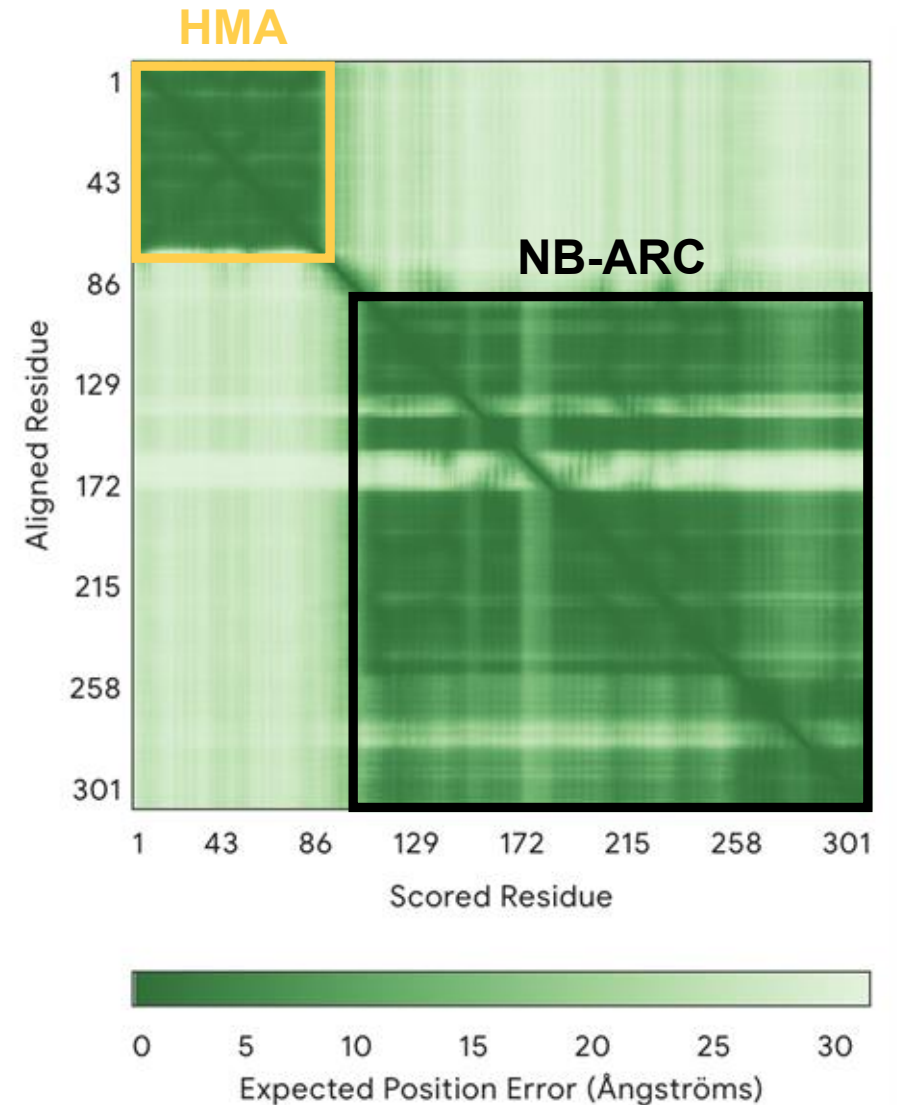
- a) What does pTM stand for? What is the pTM score for the prediction?
- b) What can you say about the local measure of confidence (pLDDT)? Is it high all over the model? Are there any amino acids predicted with higher confidence than others?
- c) What can you say about the predicted aligned error (PAE) plot? Are there any regions where the relative positions are unreliable?

Domain architecture and PAE plot of Pikp-1

Residues 186–486



Domain annotation from InterProScan



EXERCISE 2

Modelling a multimeric complex

In this exercise, you will explore the oligomeric state predictions for a given protein. You will model the HMA domain of the Pikp-1 protein (from Exercise 1.1) in **dimeric (n=2)**, **trimeric (n=3)**, and **tetrameric (n=4)** states.

Your task is to evaluate the confidence in the relative orientation of the modeled polypeptide chains by interpreting the PAE plots. You will complete the table with predicted pTM and ipTM scores and, based on this, determine the likely oligomeric state (monomer, dimer, trimer, or tetramer).

>Pikp-HMA

GLKQKIVIKVAMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLI SALRKKVGD AELLQVSQANKD

EXERCISE 2

Complete the results table

a) What does ipTM stand for?

b) Complete the table below with your modelling results:

Oligomeric state	ipTM	pTM
Monomer (n = 1)	—	
Dimer (n = 2)		
Trimer (n = 3)		
Tetramer (n = 4)		

Fill in during the session as you run each stoichiometry (n = 1 to 4).

Interpreting the PAE across oligomeric states

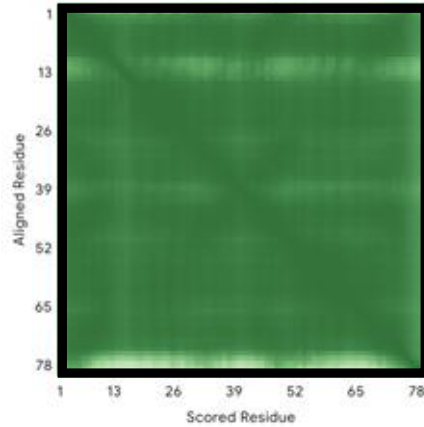


DISCUSSION

- c) What does the predicted alignment error (PAE) look like for the different oligomeric states? Is there high confidence in the relative positioning of the protomers, across all cases tested?

Pikp-HMA shows the highest ipTM score as a homodimer

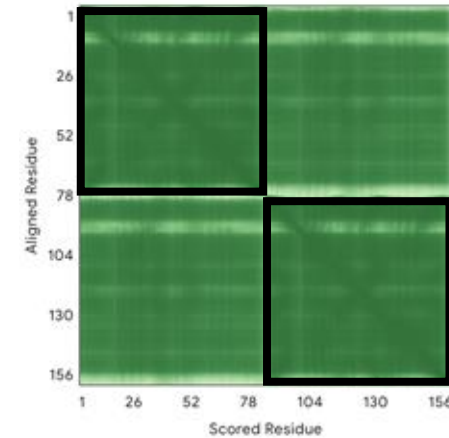
Monomer (n=1)



pTM: 0.81

ipTM: N/A

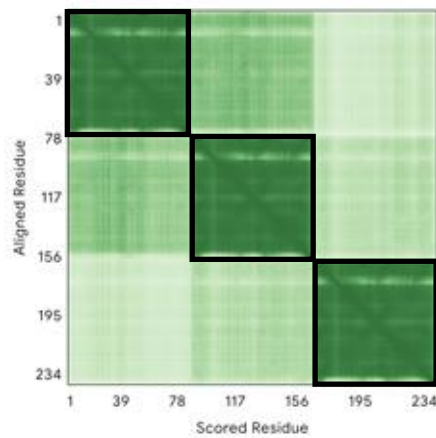
Dimer (n=2)



pTM: 0.86

ipTM: 0.83

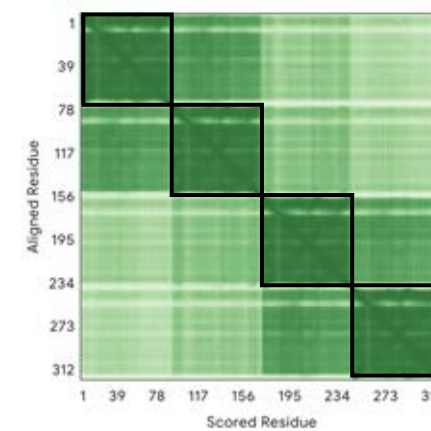
Trimer (n=3)



pTM: 0.49

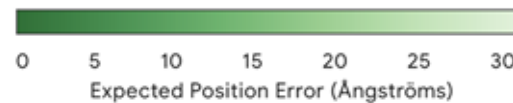
ipTM: 0.3

Tetramer (n=4)

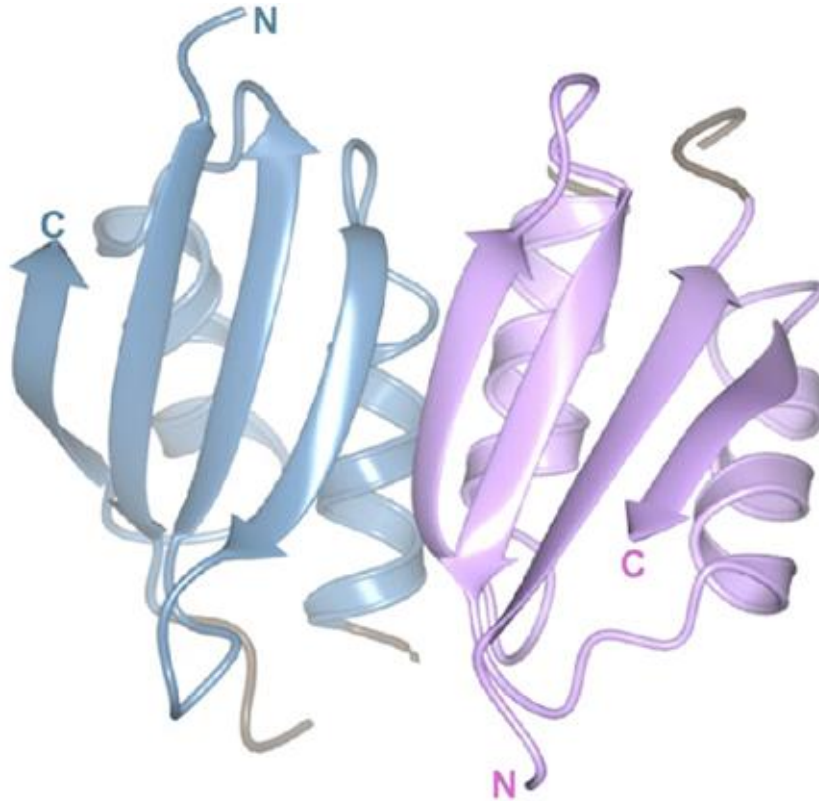


pTM: 0.59

ipTM: 0.49



Crystal structure of Pkcp-HMA has been solved as a homo-dimer



PDB: 5a6p

(Maqbool et al., 2015, PMID: 26304198)

EXERCISE 3

Protein–protein interaction prediction

Next, model three different heteromeric complexes between pathogen proteins and the rice Pikp-1 domains. Your goal is to analyze the AF3 models and make a hypothesis about **which effector may interact with the plant host protein**, based on the modelling confidence metrics alone.

>Pikp-1_186-486

```
GLKQKIVIKVAMEGNNCRSKAMALVASTGGVDSVALVGDLRDKIEVVGYGIDPIKLISALRKKVGDAELLQVSQANKDVKETTPMLAPVKSICEFHKVKTVCILGLPGGGKTTVARELYDAL  
GTHFPCR VFVS VSPSSSPNLTKTLADIFAQAQLGVTDTLSTSYGGSGTGRALQQHLIDNISAFLLNKKYLIVIDDIWHWEWEVIRKSI PKNDLGGRIIMTTRLNSIAEKCHTDDNDVFV  
YEVGDLDDNDALSLSWG IATKSGAGNRIGTGEDNPCYDIVNMCYGMPLALIWLSSAL
```

EXERCISE 3

Three pathogen effector candidates

Amino acid sequences of three pathogen proteins secreted by the fungus *Pyricularia oryzae* (causal agent of rice blast disease):

>Pathogen_protein_1 (AVR-PikD)

ETGNKYIEKRAIDL SRERDPNFFDH PGIPVPECFWFMFKNNVRQDAGTCYSSWKMDMKVGPNWVHIKSDDNCNLSGDFPPGWIVLGKKRPGF

>Pathogen_protein_2 (AVR-Pib)

ATQVTILKKGERITWVEVPKGESREFNIRGKYFTVSVSDDGTPSISGSKYTVE

>Pathogen_protein_3 (AVR-Pii)

LPTPASLNGNTEVATISDVKLEARSDTTYHKCSKCGYGSDSDAYFNHKCN

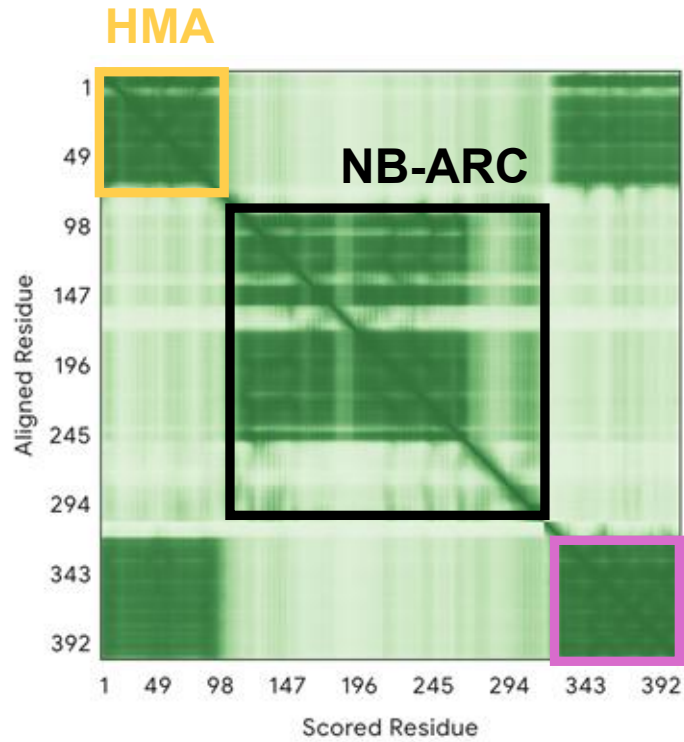
Form and test your hypothesis



DISCUSSION

- a) Based on these modelling results, which pathogen effector(s) is most likely to interact with the rice Ptkp-1? What confidence metrics allow you to conclude this?
- b) Based on the modelling results, which Ptkp-1 domain contributes to binding to the pathogen protein?
- c) What kind of experiments would you perform to test this hypothesis?

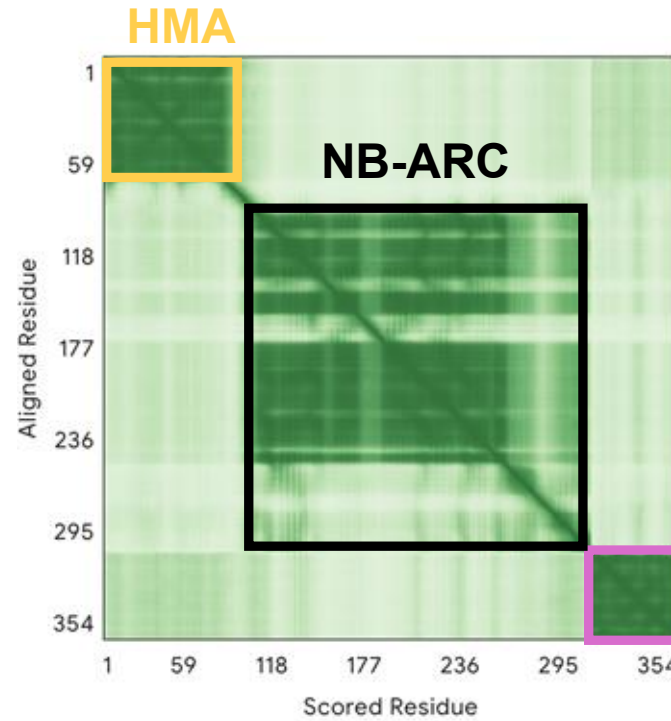
Pathogen_protein_1 shows the highest ipTM score



Pathogen_protein_1
(AVR-PikD)

pTM: 0.45

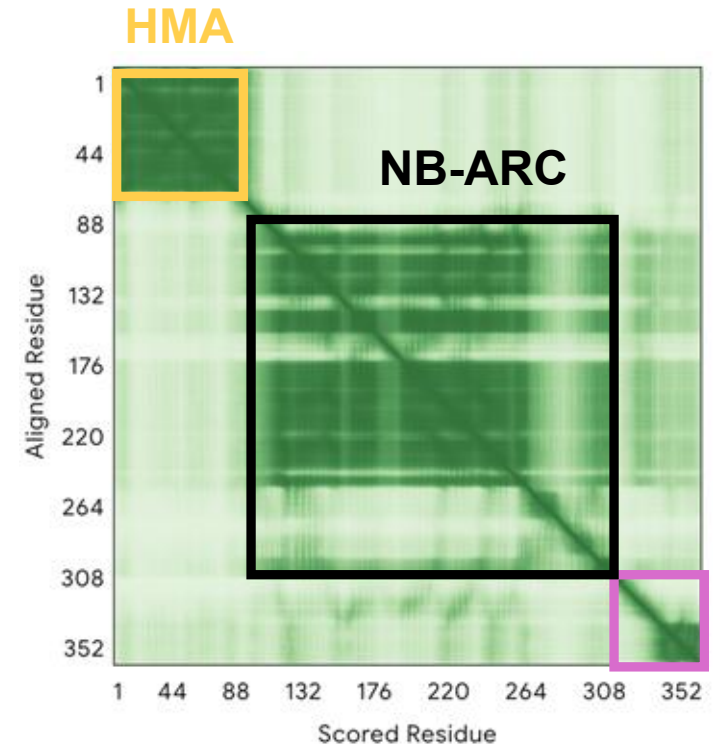
ipTM: 0.83



Pathogen_protein_2
(AVR-Pib)

pTM: 0.47

ipTM: 0.16



Pathogen_protein_3
(AVR-Pii)

pTM: 0.22

ipTM: 0.5

Programme

The same two-day programme runs at both venues (Espace Zmorda, then ISBS).

Day 1 Predict & Interpret

08:30–09:30	Registration
09:30–09:45	Welcome & Introductions
09:45–10:15	Introduction to AI for Genomics
10:15–11:00	Theoretical Introduction to AlphaFold
11:00–11:20	Break
11:20–12:30	 Practical 1: Predict
12:30–13:30	Lunch
13:30–15:00	<i>GOHREP it!</i> Interactive discussions
15:00–15:15	Break
15:15–17:00	 Practical 2: Interpret
17:00–17:15	Q&A

Day 2 Compare & Design

09:00–09:30	Day 2 Welcome & Registration
09:30–11:15	 Practical 3: Compare
11:15–11:30	Break
11:30–12:30	Model your own proteins
12:30–13:30	Lunch
13:30–14:45	 Practical 4: Design
14:45–15:05	Break
15:05–16:15	Genomics in an AI world
16:15–16:30	Summary
16:30–17:00	Q&A
17:00	Meeting close